Differential CC $1\mu1\pi^{\pm}$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

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$\nu_\mu + \bar{\nu}_\mu$ charged-current single charged pion • FHC, RHC, combined

Outline

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3. Momentum reconstruction & response
4. Wiener-SVD unfolding & p_π mitigation
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6. Integrated & total cross sections
7. Systematics & sideband validation

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3. Observables, closure & response
4. Generator comparison
5. Background composition & sidebands

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Backup

Part I: Inclusive CC $1\mu 1\pi^{\pm}Xp$

Signal definition

$\nu_\mu/\bar{\nu}_\mu$ charged-current: exactly 1μ ($p_\mu > 0.15\,\mathrm{GeV}/c$), exactly $1\pi^{\pm}$ ($p_\pi > 0.175\,\mathrm{GeV}/c$), ≥ 0 protons, no other mesons ($0\pi^0, 0K$), vertex in active fiducial volume.

- **Final-state signal**: includes primary pion production and FSI migrations.

Mode	% of signal	Pion origin
RES ($\Delta(1232)$, higher N^*)	77.5	primary
DIS / SIS	17.5	primary (higher W)
QE	2.6	FSI only (no primary π)
COH	2.1	primary, low $\lvert t\rvert$
MEC / 2p2h	0.3	FSI only

- **FSI migration**: absorption ($N\pi \to 1\pi$) and charge exchange ($1\pi \to 0\pi$).
- **NuMI beam (110 mrad)**: ν_μ-dominant FHC, $\bar{\nu}_\mu$-enriched RHC.
- **Observables**: p_μ, p_π, $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$.

Exposure & incoming flux ($E_\nu > 0.1$ GeV)

Mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$	10^{20} POT
FHC	63.4%	34.0%	1.7%	0.9%	8.86
RHC	44.2%	53.3%	1.4%	1.2%	11.08
Comb	53.0%	44.4%	1.5%	1.1%	19.94

110 mrad off-axis gives large wrong-sign flux. Signal is charge-blind ($\lvert\nu_{\mathrm{pdg}}\rvert = 14$), measuring $\nu_\mu + \bar{\nu}_\mu$ combined.

Event selection: cut sequence

Topological CC$1\mu1\pi^{\pm}Xp$ selection applied sequentially (cosmics $\rightarrow$ PID $\rightarrow$ kinematics):

1. **NeutrinoSlice**: ≥ 1 track, software trigger.
2. **InFiducialVol**: vertex inside active TPC FV.
3. **Topological**: Pandora score > 0.67 (EXT $\times 33$ down).
4. **MuonCandidate**: passes μ PID; longest track $= \mu$.
5. **ContainedPion**: 1 contained track with π PID.

6–7. μ/π **In3Planes**: hits on U, V, and Y planes.

8. **ShowerCut**: veto EM showers (π^0 rejection).
9. **OpeningAngle**: $\theta_{\mu\pi} < 2.6\,\text{rad}$ (149°) cosmic veto.
10. **Nonprotons**: < 4 non-proton tracks, < 5 total.

Final (measured phase space): efficiency 17.5%, **purity** 55.7%

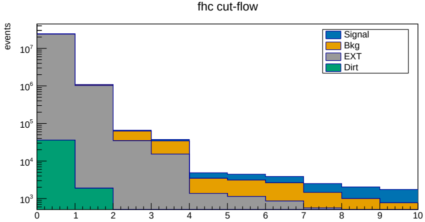
Inclusive cut-flow, FHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	5 580	279 180	23.92M	100.0	0.02
InFiducialVol	4 632	62 146	1.02M	83.0	0.42
Topological	3 501	28 040	34 481	62.7	5.2
MuonCandidate	3 143	18 980	15 206	56.3	8.3
ContainedPion	1 395	2 094	1 348	25.0	28.5
MuonIn3Planes	1 336	1 975	1 122	23.9	29.9
PionIn3Planes	1 239	1 770	854	22.2	31.9
ShowerCut	1 048	898	558	18.8	41.6
OpeningAngle	1 040	813	186	18.6	50.9
Nonprotons	**974**	**643**	**128**	**17.5**	**55.7**

MC prediction (POT-scaled). Dirt is negligible ($\lesssim 1$ event).

Final cut summary across beams:

Beam	Signal	Bkg	EXT	Eff%	Pur%
FHC	974	643	128	17.5	55.8
RHC	1 081	746	133	17.6	55.1
Comb	2 055	1 389	261	17.5	55.4



FHC stack (log scale).

Major rejections: **Topological** (cosmics $\times 33$) and **ContainedPion** (single-π topology).

Particle identification (muon & pion candidates)

Track-level PID combines Pandora track score, calorimetric LLR PID, Bragg likelihoods, and MIP/pion BDTs:

Muon candidate (Cut 4)

Variable	Threshold
Track score	> 0.80
Vertex distance	≤ 4.0 cm
Track length	> 10.0 cm
LLR PID	> 0.20
MIP BDT	≥ -0.10

Pion candidate (Cut 5)

Variable	Threshold
LLR PID	> 0.10
Vertex distance	< 4.0 cm
Track length	> 20.0 cm
MIP BDT	> -0.10
Pion BDT (vs. p)	> -0.10
Bragg-pion score	≥ 0.08
Endpoint contained	yes

Momentum reconstruction: range, MCS and proton KE

Contained tracks use range-based momentum via dE/dx; exiting muons use multiple Coulomb scattering (MCS).

Particle	Estimator	PeLEE branch	Note
μ (contained)	range momentum	`trk_range_muon_mom_v`	37% of selected
μ (exiting)	MCS momentum	`trk_mcs_muon_mom_v`	63%, range invalid
π	range momentum, μ hypothesis	`trk_range_muon_mom_v`	$m_\mu \approx m_\pi$
p	range KE, p hypothesis	`trk_energy_proton_v`	$\rightarrow$ momentum $\lvert\vec{p}_p\rvert$

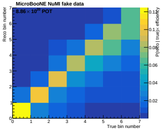
p_μ is a hybrid observable; MCS-dominated because most energetic muons leave the active volume.

Pion range momentum (μ hypothesis)

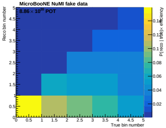
Close masses ($m_\mu = 105.7\,\text{MeV}$, $m_\pi = 139.6\,\text{MeV}$) make muon range momentum an accurate proxy for stopping pions.

Directly comparable to true p_π, preserving a near-diagonal response matrix. Requires contained pions.

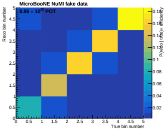
Response (smearceptance) matrices, FHC



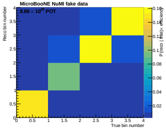
p_μ



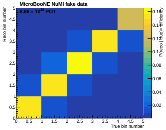
p_π



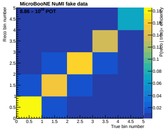
$\cos\theta_\mu$



$\cos\theta_\pi$



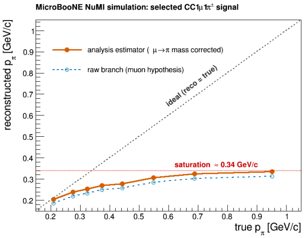
$\theta_{\mu\pi}$



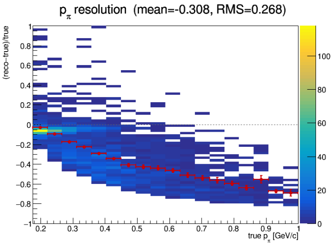
θ_μ

- Matrix elements $A_{ij} = P(\mathrm{reco}\, i \,|\, \mathrm{true}\, j) \times \varepsilon_j$; columns = true bins, rows = reco bins.
- p_μ and angular observables $(\cos\theta_\mu, \cos\theta_\pi, \theta_{\mu\pi})$ are near-diagonal; MCS resolution sets p_μ width.
- p_π exhibits severe migration: hadronic re-interactions cause premature stopping and momentum underestimation, motivating Wiener-SVD regularisation with additional smearing A_C.

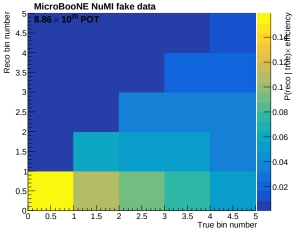
Why the pion momentum cannot be measured well



Saturates above $\sim 0.35\,\mathrm{GeV}/c$



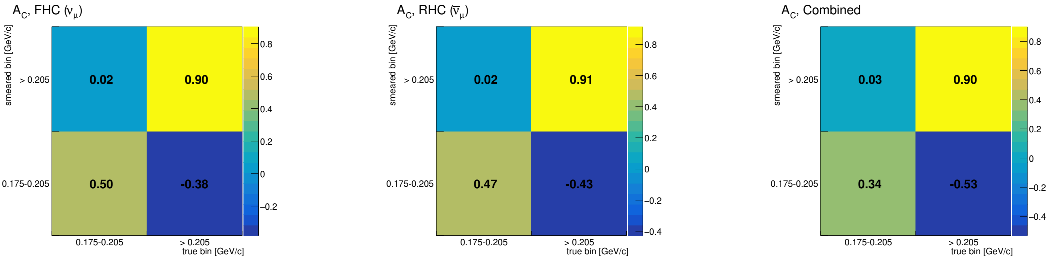
Systematic under-reconstruction



Reco piles into lowest bin

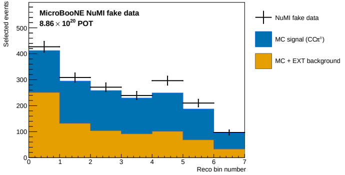
- **Hadronic re-interactions**: Unlike muons, charged pions scatter or get absorbed in argon before stopping $\Rightarrow$ premature track endings cause severe momentum underestimation.
- **Fundamental detector limit**: Tagging cleanly stopping pions via BDT (or MC truth) reaches a ceiling without resolving the migration.
- **Mitigation**: Pion momentum is reported in a robust two-bin scheme; all other kinematic observables remain unaffected.

Pion measurement reduced to two bins

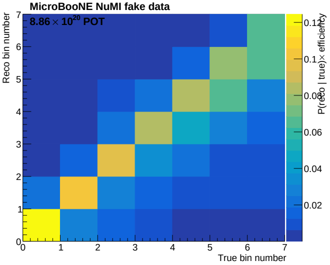


- **Inter-bin correlation**: A_C carries negative off-diagonal entries (borrowing between bins) due to strong bin-to-bin migration.
- **Configuration consistency**: FHC and RHC regularise similarly (diagonals $0.50, 0.47$); combined fit shares more weight (diagonal 0.34, off-diagonal -0.53).
- **Normalisation**: Unfolded/truth ratio is 1.13 across all three modes: an overall scale offset independent of the smearing strength of A_C.

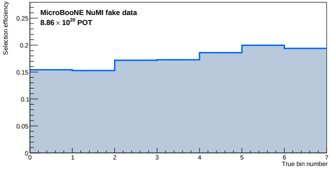
Reconstruction, response & efficiency



Reco p_μ (data/MC stack)



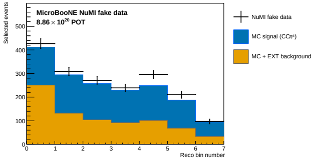
Response matrix



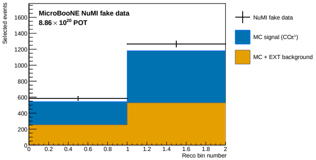
Efficiency vs p_μ

Wiener-SVD unfolding (framework `WienerSVDUnfolder`, 2nd-derivative regularisation) with full universe-throw systematic covariance.

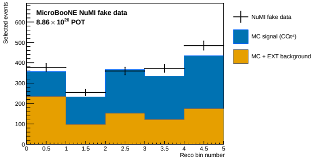
FHC: selected events (POT-scaled), signal vs background



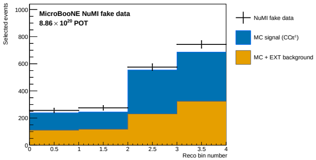
p_μ



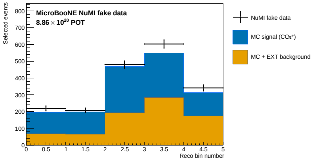
p_π



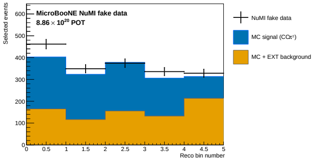
$\cos\theta_\mu$



$\cos\theta_\pi$



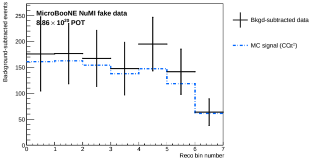
$\theta_{\mu\pi}$



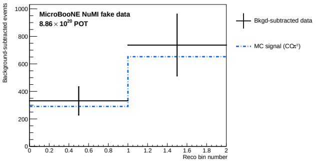
θ_μ

Reco kinematic distributions before background subtraction. Points: (blind) fake data; Stack: MC signal + background (55.7% purity).

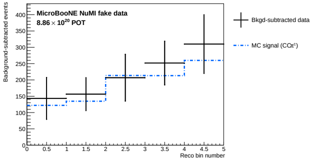
FHC: after background subtraction



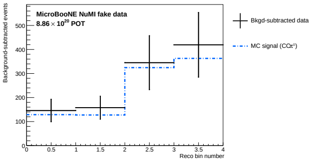
p_μ



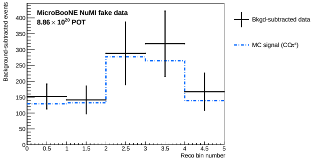
p_π



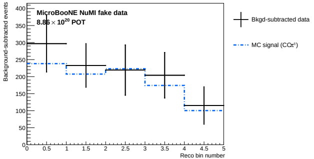
$\cos\theta_\mu$



$\cos\theta_\pi$



$\theta_{\mu\pi}$

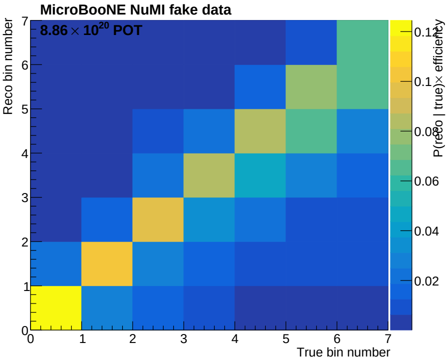


θ_μ

Data minus estimated background vs CV MC signal (unfolding input). Backgrounds from MC (ν), EXT cosmics, and dirt overlay; validated by sidebands.

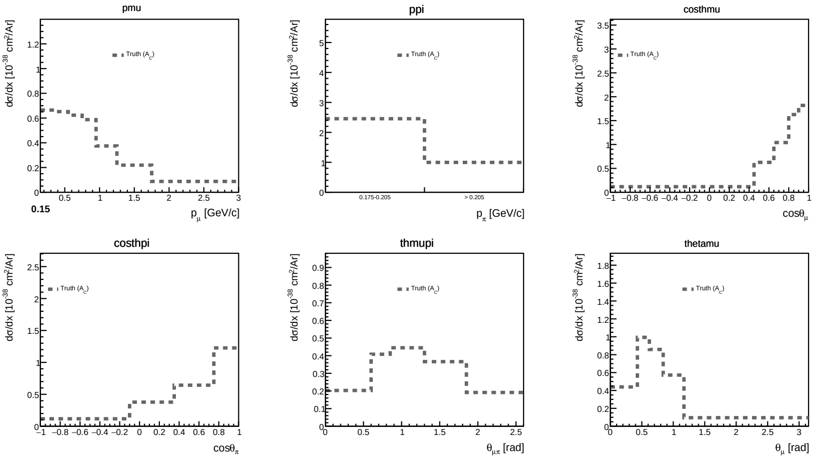
FHC: unfolding

- **Detector corrections**: Background-subtracted data still carries detector response and selection efficiency.
- **Wiener-SVD regularisation**: Inverts the response matrix while suppressing noise from 50–99% bin migrations.
- **Additional smearing (A_C)**: Unfolding outputs $A_C \times \sigma_{\mathrm{true}}$. Theoretical models are multiplied by A_C before comparing to data.
- **Progression**: Next slides build the model comparison curve by curve.



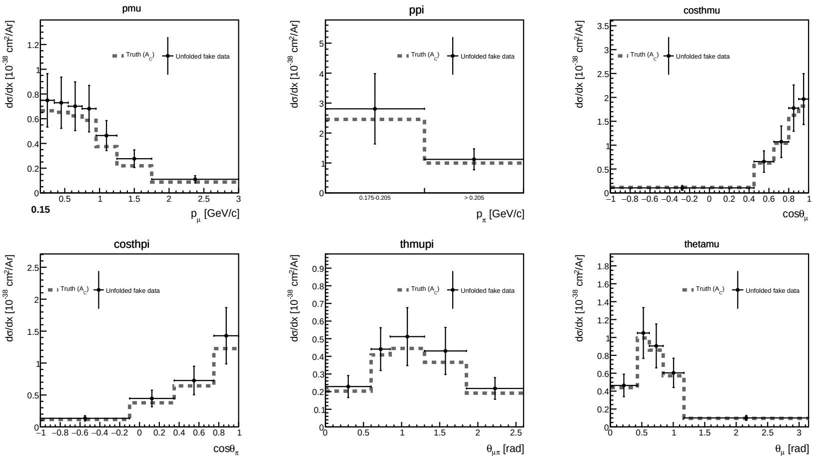
Response matrix, p_μ

recover



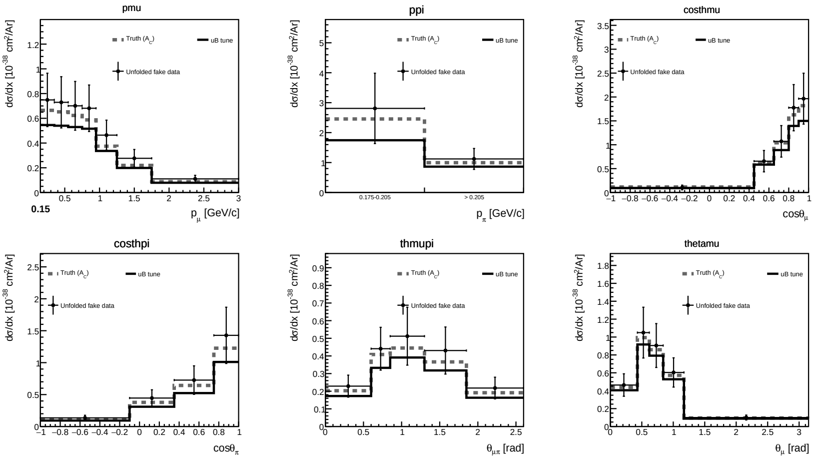
Poisson-thrown fake data generated from the MicroBooNE tune; after unfolding we should get this A_C smeared truth. p_π uses the two-bin scheme (bins: 0.030 and 0.795 GeV$/c$).

Differential cross sections, FHC: + unfolded fake data



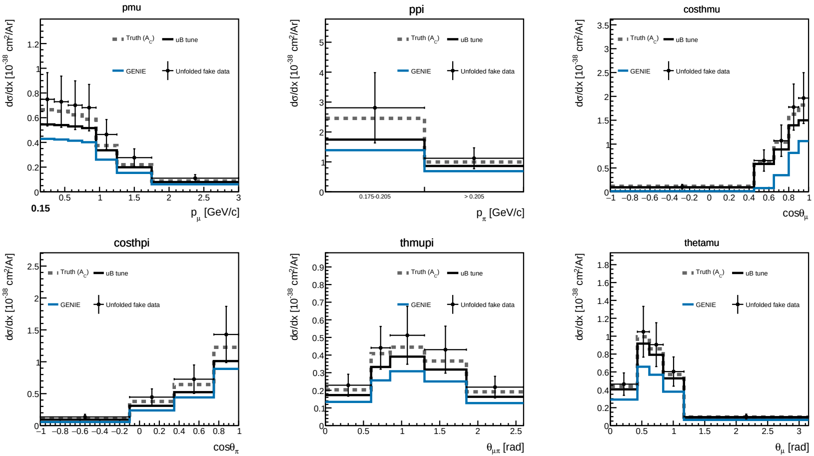
Unfolded fake data close cleanly on the A_C-smeared truth across all kinematic variables.

Differential cross sections, FHC: + MicroBooNE tune



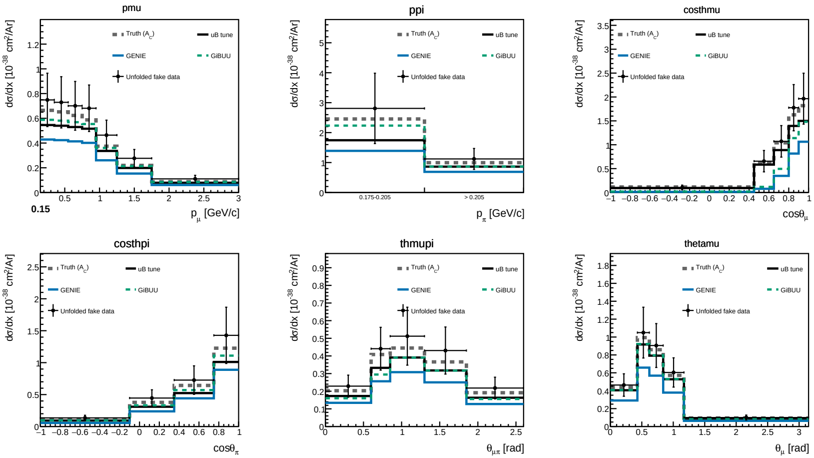
Central-value MicroBooNE tune (A_C-smeared), the model from which the fake data were thrown.

Differential cross sections, FHC: + GENIE



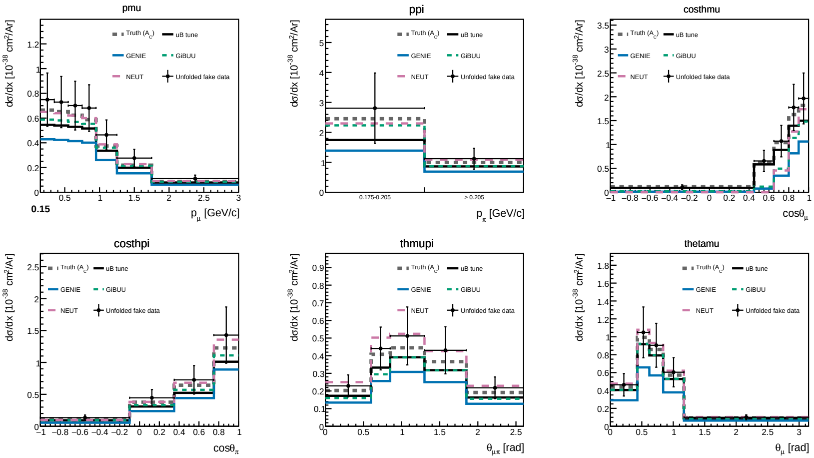
GENIE v3.0.6 (Empirical MEC + *updated* RES, A_C-smeared): captures the overall scale, slightly underpredicts forward μ. This is partially due to the A_C overcorrection - the θ_μ doesn't show such a large difference.

Differential cross sections, FHC: + GiBUU



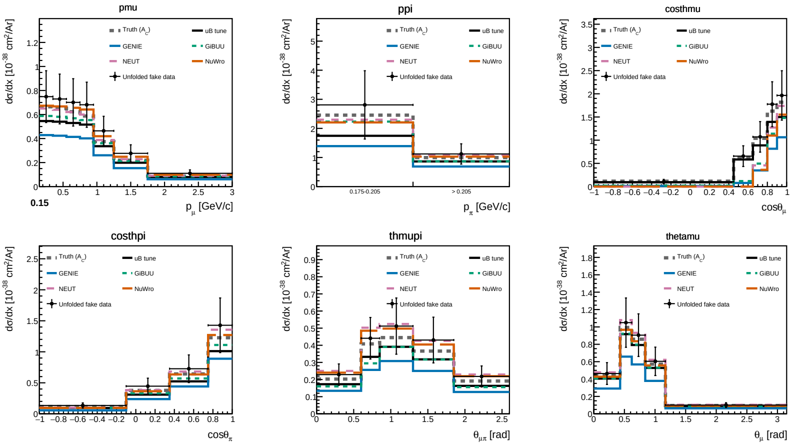
GiBUU 2023 (coupled-channel transport, A_C-smeared): detailed nuclear cascade predicts lower pion yield at forward angles.

Differential cross sections, FHC: + NEUT



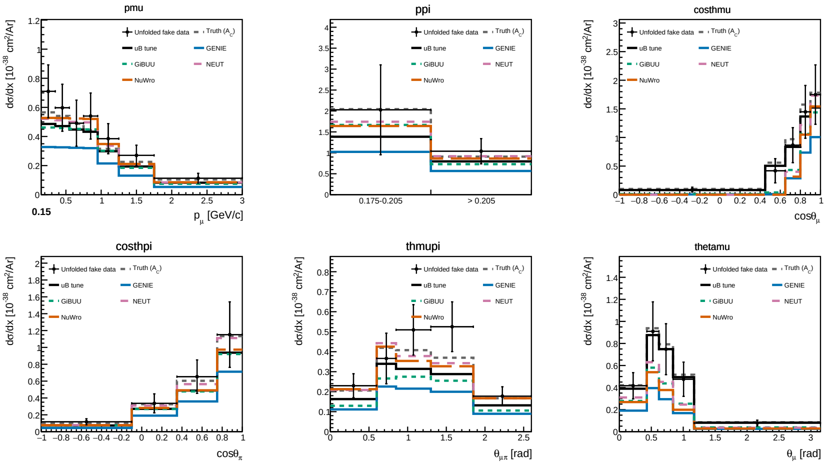
NEUT 5.4.0 (Spectral Function, A_C-smeared): predicts a more pronounced forward muon peak in $\cos\theta_\mu$- not visible in θ_μ.

Differential cross sections, FHC: + NuWro



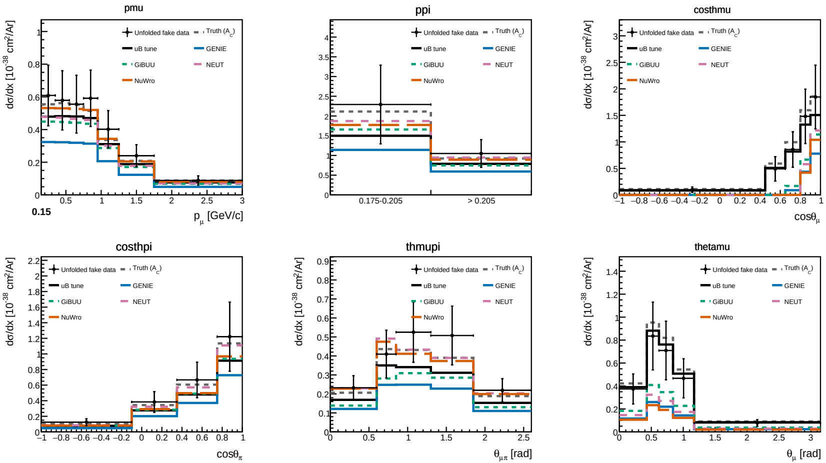
All four generators bracket the unfolded data. Error bars are stat $\oplus$ syst: full covariance propagated through unfolding ($\sim$27% total for p_μ, flux-dominated at 23%).

Differential cross sections, RHC



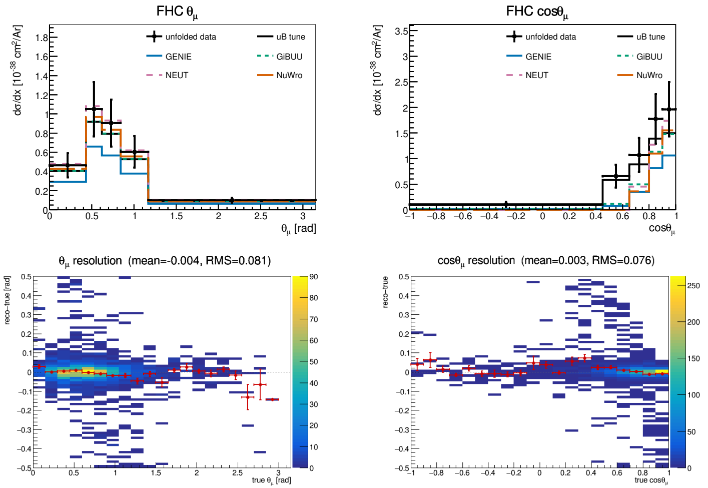
$\bar{\nu}_\mu$-enriched beam; unfolded fake data vs A_C-smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

Differential cross sections, combined



FHC+RHC combined exposure (per-universe correlated flux treatment). Error bars: stat $\oplus$ syst (full covariance).

Muon angle: $\cos\theta_\mu$ vs θ_μ



All curves A_C-smeared. Same events and binning (θ edges = arccos of $\cos\theta$ edges). The steep forward peak in $\cos\theta_\mu$ induces A_C smearing sensitivity in χ^2; θ_μ provides a more uniform, linear metric for model comparisons (detailed in backup).

Integrated & total flux-averaged cross section

Integrated σ_{int} (Wiener-SVD)

	FHC	RHC	Comb
p_μ	0.99	0.86	0.82
p_π	0.96	0.97	0.96
$\cos\theta_\mu$	0.82	0.66	0.70
$\cos\theta_\pi$	0.97	0.81	0.86
$\theta_{\mu\pi}$	0.88	0.88	0.92
θ_μ	1.00	0.83	0.78
θ_π	0.97	0.80	0.86

Spread reflects Wiener-SVD A_C smearing (10^{-38} cm^2/Ar)

Total (cut-and-count, no unfolding)

Config	σ (10^{-38} cm^2/Ar)
FHC	1.06 ± 0.27
RHC	0.98 ± 0.35
Combined	1.02 ± 0.30

Uncertainties: syst $\oplus$ stat (FHC: 25.5% syst, 4.3% data stat).

Cut-and-count carries no A_C, directly comparable to truth models.

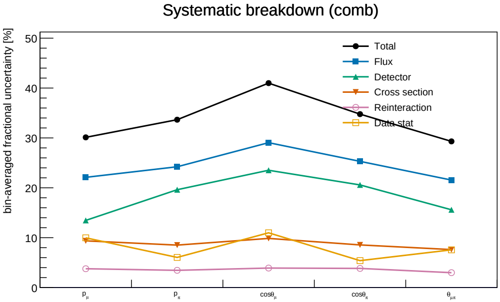
Generator totals: spread is genuine physics

	truth-level	A_C-smeared	A_C factor
GENIE	0.82	0.56	0.69
GiBUU	1.13	0.78	0.69
NEUT	1.32	0.84	0.64
NuWro	1.25	0.90	0.72

A_C applies a common $\simeq 0.7$ factor across all four models
The $1.6\times$ model spread is genuine physics.

FHC cut-and-count data: 1.06.

Systematics & sideband validation



PPFX beam flux dominates (22–29%); detector modeling second; total $\sim$ 30–40%.

Background-control sidebands (combined)

Region	N_{data}	EXT	sig%	data/νMC
CC0π	24 148	17 855	9.7	1.00
π^0	14 694	10 284	9.9	1.01
multi-π	303	83	22.9	1.08
cosmic	169	616	8.0	1.10

Inverting single cuts tests key modeling components:
CC0π (FSI absorption),
π^0 (shower veto),
multi-π (pion cascade),
cosmic (EXT-dominated).

Data/νMC $\approx$ 1.00 validates POT scaling and weighting pipeline before unblinding.
Sidebands will constrain backgrounds to $\sim$1%.
Three beam configurations will additionally help to constrain the background.

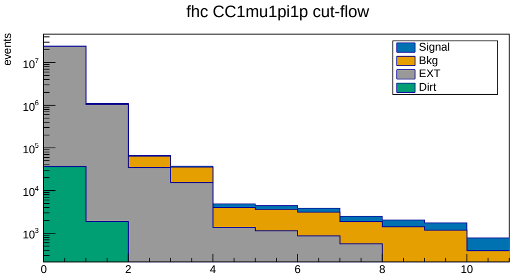
Part II: Proton-tagged subsample (W / TKI)

Physics motivation

Requiring an identified proton accesses the complete hadronic final state, unlocking hadronic invariant mass (W, probing Δ/N^* resonances) and transverse kinematic imbalance (TKI, probing Fermi motion and FSI).

- **Selection**: Inherits inclusive cuts + leading contained proton (LLR PID, $p_p \geq 0.3\,\text{GeV}/c$). Efficiency: 11.5%, purity: 48.4%.
- **11 Observables**: 6 W/TKI ($W_{\pi p}, W_{\text{had}}, \delta p_T, \delta\alpha_T, \delta\phi_T, p_n$) in 6 bins each, plus 5 inclusive kinematic observables re-evaluated on the proton-tagged phase space.

Selection performance & proton-tag quality



FHC proton-tagged cut-flow (last stage: leading-proton ID)

Proton tag metrics (FHC truth-matched)

- **Tag purity**: 87% (13% mis-tag rate)
- **Bkg rejection**: cuts 64% of ν background for 30% signal loss
- **Sample purity**: 33% $\rightarrow$ 48% ($S/\sqrt{B}$ +17%)

Proton-tagged cut-flow, FHC (full exposure)

Stages 1–10 inherit the inclusive selection; the final **IdentProton** cut requires a leading contained proton ($p_p > 0.3\,\mathrm{GeV}/c$, LLR PID).

Cut	Signal	Bkg	EXT	Pred	Eff%	Pur%
None	3 387	281 372	23.92M	24.7M	100.0	0.01
InFiducialVol	2 834	63 944	1.02M	1.11M	83.7	0.26
Topological	2 238	29 303	34 481	66 929	66.1	3.3
MuonCandidate	2 031	20 092	15 206	37 776	60.0	5.4
ContainedPion	852	2 637	1 348	4 890	25.2	17.4
MuonIn3Planes	812	2 499	1 122	4 473	24.0	18.2
PionIn3Planes	754	2 255	854	3 888	22.3	19.4
ShowerCut	632	1 313	558	2 521	18.7	25.1
OpeningAngle	628	1 226	186	2 045	18.5	30.7
Nonprotons	568	1 049	128	1 749	16.8	32.5
IdentProton	**401**	**373**	**46**	**828**	**11.8**	**48.4**

IdentProton boosts sample purity from 32.5% → 48.4%. RHC and combined achieve 46.6% and 47.5% purity at ∼12% efficiency.

The six proton-tagged observables

Constructed from μ, $\pi^{\pm}$, and leading p four-vectors ($\vec{p}^{\,\mathrm{had}} = \vec{p}_\pi + \vec{p}_p$; transverse components w.r.t. beam $\hat{z}$):

Hadronic mass (probes Δ/N^*)

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}$$
$$W_{\mathrm{had}} = \sqrt{m_p^2 + 2m_p(E_{\mathrm{cal}} - E_\mu) - Q^2}$$

$W_{\pi p}$ is directly measured (no ν beam assumption). W_{had} is calorimetric ($E_{\mathrm{cal}} = E_\mu + E_\pi + T_p$).
Physical threshold: $W_{\pi p} \geq m_\pi + m_p = 1.08\,\mathrm{GeV}/c^2$.

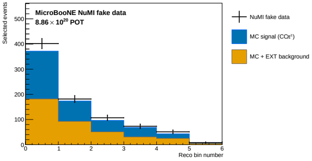
TKI (probes Fermi motion & FSI)

$$\delta p_T = |\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}}|$$
$$\delta\alpha_T = \angle(-\vec{p}_T^{\,\mu},\ \vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}})$$
$$\delta\phi_T = \angle(-\vec{p}_T^{\,\mu},\ \vec{p}_T^{\,\mathrm{had}})$$
$$p_n = \sqrt{\delta p_T^2 + p_L^2}$$

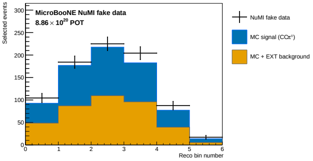
$\delta p_T \to 0$ without nuclear effects; Fermi motion broadens δp_T and FSI shifts $\delta\alpha_T \to 180^\circ$. p_n infers initial nucleon momentum ($E_b = 24.78\,\mathrm{MeV}$ argon binding energy).

Measured in six equal-population bins. Requires contained proton with $p_p > 0.3\,\mathrm{GeV}/c$.

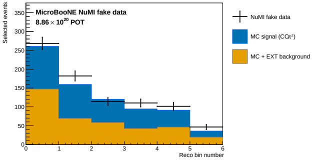
Proton-tagged, FHC: selected events (POT-scaled)



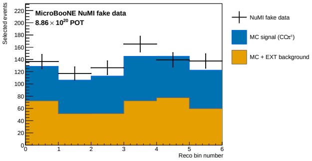
$W_{\pi p}$



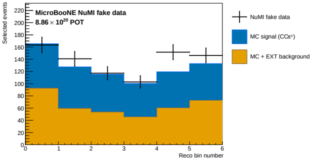
W_{had}



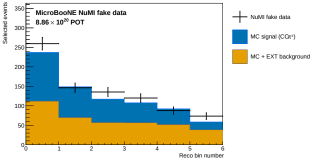
δp_T



$\delta\alpha_T$



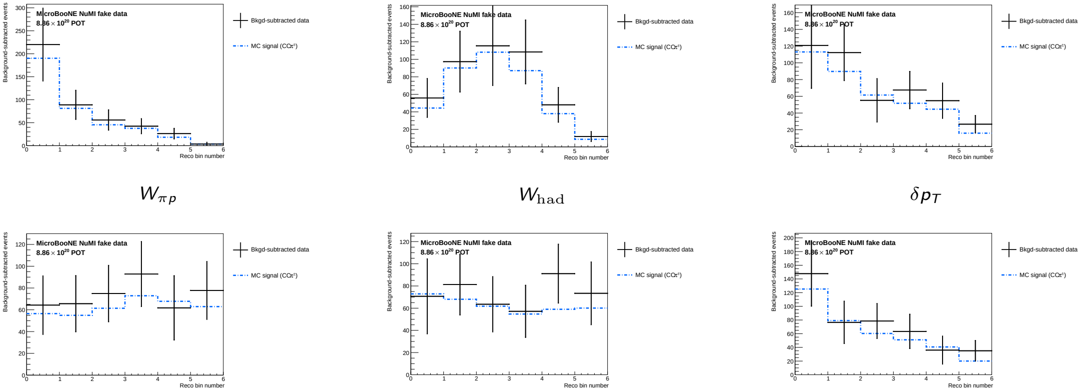
$\delta\phi_T$



p_n

Points: (blind) fake data; stack: MC signal + background. Sample purity is 48.4% at the final cut.

Proton-tagged, FHC: after background subtraction

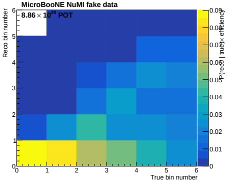


$W_{\pi p}$ W_{had} δp_T

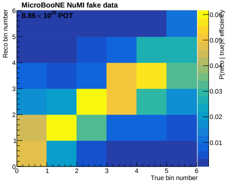
$\delta\alpha_T$ $\delta\phi_T$ p_n

Data minus estimated background vs CV MC signal (unfolding input for W/TKI observables).

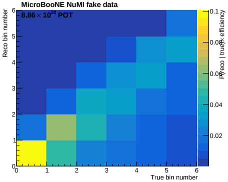
Proton-tagged: response (smearceptance) matrices, FHC



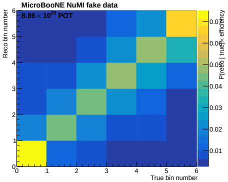
$W_{\pi p}$



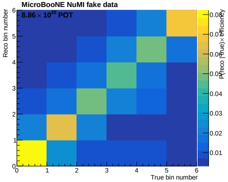
W_{had}



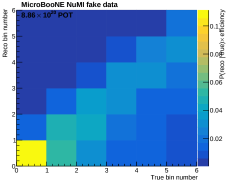
δp_T



$\delta\alpha_T$



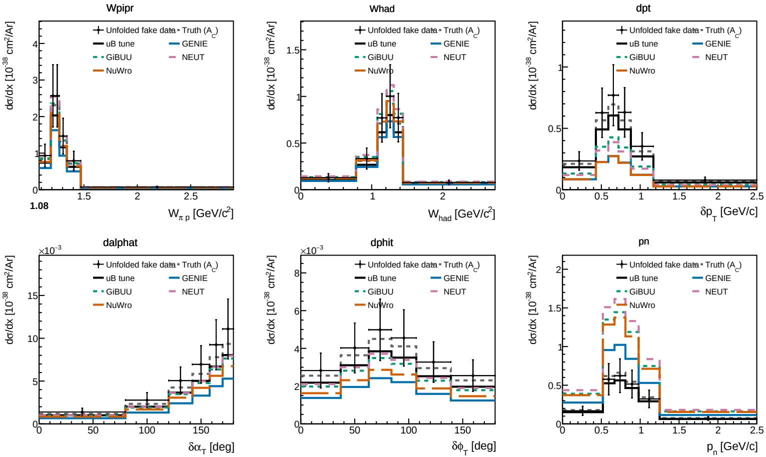
$\delta\phi_T$



p_n

A_C smearing: W_{had} (0.88), $W_{\pi p}$ (0.78), $\delta\alpha_T$ (0.63), $\delta\phi_T$ (0.61), δp_T (0.43), p_n (1.71). χ^2 tracks $|A_C - 1|$ (backup).

Proton-tagged differential cross sections (FHC)



Unfolded fake data vs A_C-smeared truth, MicroBooNE tune, and 4 external generators. Excellent closure ($\chi^2/\mathrm{ndf} \ll 1$) for every observable.

Proton-tagged closure: all three beam configurations

Observable	$\sigma_{\rm int}$ [10^{-38}cm^2/Ar] FHC	RHC	COMB	COMB χ^2/ndf
$W_{\pi p}$	0.62	0.578	0.564	0.07/6
$W_{\rm had}$	0.60	0.500	0.504	1.30/6
δp_T	0.61	0.493	0.497	0.71/6
$\delta\alpha_T$	0.67	0.567	0.667	0.86/6
$\delta\phi_T$	0.63	0.568	0.573	0.05/6
p_n	0.52	0.548	0.556	1.00/6
p_μ	0.56	0.531	0.523	0.65/7
p_π	0.60	0.572	0.582	2.49/5
$\cos\theta_\mu$	0.52	0.512	0.529	0.52/5
$\cos\theta_\pi$	0.54	0.444	0.454	2.17/4
$\theta_{\mu\pi}$	0.50	0.410	0.447	0.21/5

Closure summary (all beams)

- **Unbiased recovery**: Unfolding recovers A_C-smeared truth with $\chi^2/\mathrm{ndf} \ll 1$ across all 11 observables.
- **Positive definite**: Cross sections remain strictly positive in every bin.
- **Flux consistency**: RHC cross sections sit $\sim$10–15% below FHC, consistent with the $\bar{\nu}_\mu$ flux composition.
- **Robust regularisation**: Wiener-SVD avoids oscillatory artefacts across both kinematics and W/TKI variables.

Generator comparison & background composition

Integrated proton-tagged σ ($10^{-38}\text{cm}^2/\text{Ar}$)

Model	σ (truth level)
GENIE	0.50
NuWro	0.65
GiBUU	0.74
NEUT	0.78
Unfolded data ($W_{\pi p}$)	0.62

Spread $\sim$1.6$\times$, largest in low-δp_T / high-p_n (FSI-sensitive region). Generator values are truth-level (no A_C).

Remaining background (FHC)

Component	%
ν_μ CC wrong topology	64
True CC1μ1π, no p>0.3	18
NC	11
Out-of-FV	5
ν_e CC	1

82% true ν_μ CC; well-modeled and constrainable by sidebands.

Proton-tagged: background-control sidebands

Inherits all four sidebands from the inclusive selection without reprocessing; evaluated with and without the tagged proton requirement.

Region (comb)	as inherited				+ proton tag			
	N_{data}	EXT	sig%	d/νMC	N_{data}	EXT	sig%	d/νMC
CC0π	24 148	17 855	9.7	1.00	16 555	2 891	6.0	1.01
π^0	14 694	10 284	9.9	1.01	9 019	2 370	7.1	1.02
multi-π	303	83	22.9	1.08	137	36	18.9	1.07
cosmic	169	616	8.0	1.10	60	12	7.1	1.07

- High signal depletion (6–7% signal vs 48% in signal region).
- Proton tag suppresses EXT cosmics by $\sim 6\times$, reducing cosmic normalisation uncertainty.
- Data/νMC $\approx$ 1.00 validates event weighting across regions.
- An inverted-PID region will independently constrain the 13% proton mis-tag rate.

Part III: Multi-pion (exclusive CC $1\mu2\pi$ and CC $1\mu3\pi$)

Exclusive multi-pion channels probing the resonant/DIS transition and intranuclear pion cascade:

Tuning vs. 1π selection

Setting	1π	Multi-π
Pion–vertex distance	$\leq$ 4 cm	$\leq$ 9.5 cm (looser)
Uncontained pions	0	up to 2
Opening-angle cut	on	off
Shower veto	on	off
Wire-gap cuts	on	off
Per-pion threshold	0.175	0.10 GeV/c
Pion ID	LLR	dedicated BDT

Performance (Run 1 FHC only)

	2π	3π
Efficiency	17.9%	10.1%
Purity	26.8%	11.3%
Events @ full exposure	$\sim$670	$\sim$66
Reco ceiling	66.6%	52.4%

Efficiency/purity evaluated on Run-1 FHC ν_μ overlay; event yields extrapolated to full exposure.

The reco ceiling (reconstructed track efficiency) shows losses stem from particle-ID, not tracking.

The work is based on the investigations done by Tusharadri and Linhui.

Multi-pion: dedicated pion-ID BDT

- **Gradient BDT** (ROC $= 0.81$): separates true pions from protons, muons, and mis-reconstructed tracks in multi-track events.
- **Inputs**: LLR PID, Bragg likelihoods, track score, length, vertex distance.
- **Multiplicity strategy**: 2π uses two momentum-specialised BDTs (split at 20 cm length); 3π uses a single soft-pion BDT.
- **Topology specific**: BDT improves multi-π performance; inclusive selection retains standard LLR PID.

Residual background (truth-matched)

Counted "pion" is	%
Out-of-topology $\pi^{\pm}$	44
Proton mis-ID	31
Muon	12
Other	13

Proton mis-ID is the irreducible core (ranging protons mimic pions).

Target: differential σ for 2π, total σ for 3π.

Summary

- Inclusive CC1μ1$\pi^{\pm}$*Xp* (FHC, RHC, Combined): Differential cross sections unfolded via Wiener-SVD. Total flux-averaged σ agrees with GENIE, GiBUU, NEUT, and NuWro within $\sim 1\sigma$. Three beam modes provide independent consistency checks.
- Proton-tagged subsample: First measurement of hadronic mass ($W_{\pi p}, W_{\mathrm{had}}$) and TKI ($\delta p_T, \delta\alpha_T, \delta\phi_T, p_n$). Unfolding closure validated across all 11 observables in all three beam configurations ($\chi^2/\mathrm{ndf} < 1$).
- Data-driven sidebands: Four dedicated sidebands (CC0π, multi-π, π^0, cosmic) achieve $\mathrm{fakedata}/\nu\mathrm{MC} \approx 1.00$.
- Multi-pion: Exclusive CC1μ2π and CC1μ3π channels developed with a dedicated gradient BDT (ROC $= 0.81$). More work to do.

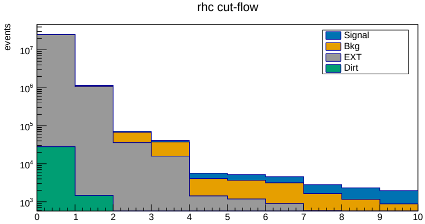
Thank you — Questions?

Backup

Backup: inclusive cut-flow, RHC (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	6 131	304 347	24.84M	100.0	0.02
InFiducialVol	5 124	69 102	1.06M	83.6	0.44
Topological	3 960	31 629	35 801	64.6	5.5
MuonCandidate	3 559	21 443	15 788	58.0	8.6
ContainedPion	1 577	2 649	1 400	25.7	27.8
MuonIn3Planes	1 512	2 500	1 165	24.7	29.0
PionIn3Planes	1 409	2 258	887	23.0	30.8
ShowerCut	1 171	1 061	579	19.1	41.4
OpeningAngle	1 162	965	193	19.0	50.0
Nonprotons	**1 081**	**746**	**133**	**17.6**	**55.0**

RHC (Runs 1,2,3,4a-c; 1.108×10^{21} POT). MC prediction, per-run POT scaled; data blind.

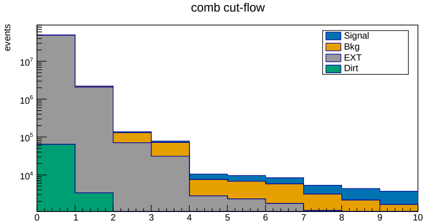


RHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale).

Backup: inclusive cut-flow, combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	11 710	583 527	48.76M	100.0	0.02
InFiducialVol	9 756	131 249	2.08M	83.3	0.43
Topological	7 461	59 669	70 282	63.7	5.4
MuonCandidate	6 702	40 423	30 994	57.2	8.5
ContainedPion	2 972	4 743	2 749	25.4	28.2
MuonIn3Planes	2 847	4 475	2 287	24.3	29.4
PionIn3Planes	2 648	4 028	1 742	22.6	31.3
ShowerCut	2 218	1 959	1 137	18.9	41.5
OpeningAngle	2 202	1 778	379	18.8	50.4
Nonprotons	**2 055**	**1 389**	**261**	**17.5**	**55.4**

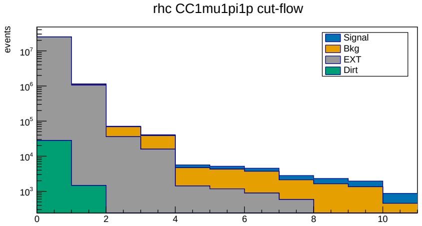
Combined (1.994×10^{21} POT). Efficiency/purity track the two beams closely; the selection is beam-independent.



Combined stack (log scale).

Backup: proton-tagged cut-flow, RHC (full exposure)

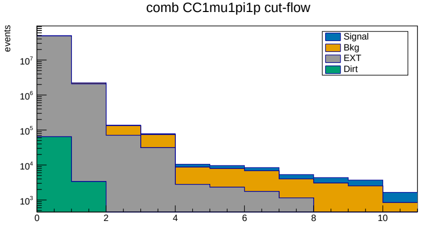
Cut	Signal	Bkg	EXT	Eff%	Pur%
None	3 638	306 840	24.84M	100.0	0.01
InFiducialVol	3 064	71 162	1.06M	84.2	0.26
Topological	2 455	33 134	35 801	67.5	3.4
MuonCandidate	2 234	22 769	15 788	61.4	5.4
ContainedPion	935	3 291	1 400	25.7	16.5
MuonIn3Planes	892	3 119	1 165	24.5	17.1
PionIn3Planes	831	2 835	887	22.9	18.2
ShowerCut	687	1 544	579	18.9	24.3
OpeningAngle	683	1 444	193	18.8	29.4
Nonprotons	611	1 217	133	16.8	31.1
IdentProton	**436**	**443**	**48**	**12.0**	**46.6**



RHC proton-tagged stack; last stage IdentProton (leading-proton tag). Purity 31.1% → 46.6%.

Backup: proton-tagged cut-flow, combined (full exposure)

Cut	Signal	Bkg	EXT	Eff%	Pur%
None	7 025	588 212	48.76M	100.0	0.01
InFiducialVol	5 898	135 106	2.08M	84.0	0.26
Topological	4 693	62 436	70 282	66.8	3.4
MuonCandidate	4 265	42 860	30 994	60.7	5.4
ContainedPion	1 787	5 928	2 749	25.4	16.9
MuonIn3Planes	1 704	5 618	2 287	24.3	17.6
PionIn3Planes	1 585	5 091	1 742	22.6	18.7
ShowerCut	1 320	2 857	1 137	18.8	24.7
OpeningAngle	1 310	2 670	379	18.7	30.0
Nonprotons	1 179	2 266	261	16.8	31.8
IdentProton	**837**	**816**	**95**	**11.9**	**47.5**



Combined proton-tagged stack; IdentProton purity $31.8\% \rightarrow 47.5\%$.

Backup: per-run exposure

Mode	Run	Data POT	MC/data scale
FHC	1	3.28×10^{20}	0.141
	2	1.27×10^{20}	0.051
	4	2.08×10^{20}	0.073
	5	2.23×10^{20}	0.116
RHC	1–4	1.11×10^{21}	0.045–0.091

Each run scaled by its own $D_{\mathrm{run}}/\mathrm{MC}_{\mathrm{run}}$ (real-data-ready).

Backup: systematic breakdown (combined)

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Total	30.1	33.7	41.0	34.8	29.3
Flux (PPFX)	22.1	24.2	29.1	25.3	21.6
Detector	13.5	19.6	23.5	20.6	15.6
Cross section	9.4	8.5	9.9	8.6	7.6
Data stat	10.0	6.1	11.0	5.4	7.6

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

Backup: historical note; pion-momentum estimator

An earlier iteration reconstructed p_π using the proton-hypothesis range KE (`trk_energy_proton_v`):

- **Mismatch**: Kinetic energy was mapped directly onto a momentum axis, distorting the migration matrix.
- **Consequence**: A true 0.20 GeV/c pion gave $\mathrm{KE} \approx 0.10$ GeV (shifted two bins low); low-momentum pions fell below the first reco bin.
- **Resolution**: Switching to `trk_range_muon_mom_v` exploits $m_\mu \approx m_\pi$ and restores a near-diagonal response matrix.

Note: Proton-hypothesis range KE remains the correct estimator for protons in the W/TKI subsample ($|\vec{p}_p| = \sqrt{T_p^2 + 2T_p m_p}$).

Backup: from PeLEE branches to kinematic inputs

Observables are constructed from 3 reconstructed tracks $(\mu, \pi^{\pm}, p)$ in PeLEE `NeutrinoSelectionFilter` ntuples:

Quantity	PeLEE branch	Note
$\lvert\vec{p}_\mu\rvert$	`trk_range_muon_mom_v`	range-based (contained)
$\lvert\vec{p}_\mu\rvert$ (MCS)	`trk_mcs_muon_mom_v`	multiple-scattering (exiting)
$\lvert\vec{p}_\pi\rvert$	`trk_range_muon_mom_v`	μ-hypothesis range at π index
$T_p \to \lvert\vec{p}_p\rvert$	`trk_energy_proton_v`	p-hypothesis range KE ($\lvert\vec{p}_p\rvert = \sqrt{T_p^2 + 2T_p m_p}$)
$\hat{u}_\mu, \hat{u}_\pi, \hat{u}_p$	`trk_dir_x/y/z_v`	3D unit track direction
$\cos\theta$	$\hat{u}_z$	w.r.t. detector z (near-beam) axis
$\theta_{\mu\pi}$	$\arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$	muon-pion opening angle
PID / candidacy	`trk_llr_pid_score_v`, `trk_bragg_*_v`, `trk_score_v`, `trk_distance_v`, `trk_len_v`	Pandora score, lengths, and vertex distances

Four-momenta: $E_i = \sqrt{\lvert\vec{p}_i\rvert^2 + m_i^2}$ using assigned PDG masses. Truth matching via `backtracked_pdg`.

Backup: W/TKI observable definitions

Reconstructed from μ, $\pi^\pm$, and leading p four-vectors ($\vec{p}^{\,\mathrm{had}} = \vec{p}_\pi + \vec{p}_p$):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad \text{(direct, beam-independent)}$$

$$W_{\mathrm{had}} = \sqrt{m_p^2 + 2m_p(E_{\mathrm{cal}} - E_\mu) - Q^2} \quad \text{(calorimetric)}$$

$$\delta p_T = |\vec{p}_T^{\,\mu} + \vec{p}_T^{\,\mathrm{had}}|, \quad \delta\alpha_T,\ \delta\phi_T\ \text{(TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

$E_{\mathrm{cal}} = E_\mu + E_\pi + T_p$; transverse components taken w.r.t. beam ($\hat{z}$) axis. Argon binding energy $E_b = 24.78\,\mathrm{MeV}$ enters p_L (and p_n).

Backup: proton-tagged binning & closure

Observable	Bin edges (6 equal-population bins)
$W_{\pi p}$	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.00 GeV/c^2
W_{had}	0, 0.78, 1.08, 1.21, 1.31, 1.44, 1.90 GeV/c^2
δp_T	0, 0.43, 0.59, 0.72, 0.88, 1.17, 1.60 GeV/c
$\delta \alpha_T$	0, 80, 120, 142, 158, 170, 180 deg
$\delta \phi_T$	0, 37, 63, 84, 107, 139, 180 deg
p_n	0, 0.52, 0.67, 0.81, 0.98, 1.25, 1.70 GeV/c

+ 5 inclusive kinematic observables on their standard binning.

Closure: Unbiased recovery across all 11 observables ($\chi^2/\mathrm{ndf} \ll 1$, strictly positive). RHC sits 10–15% below FHC as expected from flux composition.

A_C **smearing factors**: W_{had} (0.88), $W_{\pi p}$ (0.78), $\delta\alpha_T$ (0.63), $\delta\phi_T$ (0.61), δp_T (0.43), p_n (1.71).

Backup: χ^2/ndf, inclusive, all observables and beams

Unfolded fake data compared with each A_C-smeared prediction.

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	1.66/7	3.09/7	5.99/7	3.87/7	3.77/7	2.90/7
FHC	p_π	1.96/5	3.77/5	5.05/5	3.78/5	3.33/5	3.46/5
FHC	$\cos\theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	$\cos\theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	$\theta_{\mu\pi}$	0.59/5	2.59/5	4.12/5	3.24/5	2.47/5	2.29/5
RHC	p_μ	5.27/7	8.57/7	10.88/7	8.81/7	8.15/7	8.48/7
RHC	p_π	6.09/5	7.04/5	9.36/5	7.28/5	5.96/5	6.13/5
RHC	$\cos\theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	$\cos\theta_\pi$	1.47/4	4.00/4	6.64/4	3.92/4	4.21/4	4.30/4
RHC	$\theta_{\mu\pi}$	4.32/5	5.57/5	7.67/5	5.68/5	6.56/5	7.59/5
Comb	p_μ	3.41/7	3.70/7	5.82/7	3.99/7	3.83/7	3.30/7
Comb	p_π	1.32/5	2.32/5	3.85/5	2.65/5	1.82/5	2.05/5
Comb	$\cos\theta_\mu$	2.49/5	4.61/5	9.07/5	12.02/5	26.72/5	27.20/5
Comb	$\cos\theta_\pi$	0.42/4	1.83/4	4.01/4	1.69/4	1.89/4	1.95/4
Comb	$\theta_{\mu\pi}$	2.10/5	2.42/5	3.91/5	3.05/5	3.49/5	3.45/5

Truth column confirms excellent closure. Outlier $\cos\theta_\mu$ χ^2 is an A_C artefact (see next slide).

Backup: the $\cos\theta_\mu$ χ^2 is largely an A_C artefact

Same events and binning; only the angular representation changes (θ vs $\cos\theta$):

Beam	Variable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	$\cos\theta_\mu$	0.47/5	3.62/5	9.67/5	12.43/5	26.28/5	26.02/5
FHC	θ_μ	0.05/5	3.07/5	4.73/5	3.03/5	2.82/5	2.90/5
RHC	$\cos\theta_\mu$	0.96/5	4.91/5	7.45/5	8.66/5	18.43/5	17.42/5
RHC	θ_μ	0.48/5	4.41/5	6.97/5	5.54/5	8.18/5	8.46/5
FHC	$\cos\theta_\pi$	0.49/4	1.20/4	3.82/4	0.99/4	1.34/4	1.19/4
FHC	θ_π	0.44/4	1.15/4	3.77/4	0.93/4	1.30/4	1.18/4

Mechanism: In $\cos\theta_\mu$, A_C strongly suppresses sharply peaked forward models ($\times 0.33$), artificially inflating χ^2 for NEUT/NuWro. In θ_μ (linearized spacing), the same models agree well (χ^2 drops $\sim 9\times$).

Conclusion: The large $\cos\theta_\mu$ χ^2 is an A_C regularisation metric artefact, motivating θ_μ reporting.

Backup: χ^2/ndf, proton-tagged (FHC)

Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro	A_C
W_{had}	0.09/6	1.39/6	1.65/6	1.06/6	1.17/6	1.05/6	0.88
$W_{\pi p}$	0.09/6	0.95/6	1.78/6	0.62/6	0.56/6	0.87/6	0.78
$\delta\alpha_T$	0.26/6	2.74/6	4.73/6	2.95/6	2.78/6	3.53/6	0.63
$\delta\phi_T$	0.09/6	5.14/6	7.21/6	5.61/6	5.39/6	6.47/6	0.61
δp_T	0.46/6	5.34/6	9.89/6	8.27/6	9.02/6	10.38/6	0.43
p_n	0.09/6	4.24/6	7.72/6	18.74/6	25.40/6	16.45/6	1.71

Observables ordered by $|A_C - 1| = 0.12, 0.22, 0.37, 0.39, 0.57, 0.71$. Model χ^2 tracks $|A_C - 1|$ monotonically:

- Observables with $A_C \approx 1$ ($W_{\mathrm{had}}, W_{\pi p}$) show excellent agreement across all four generators ($\chi^2/\mathrm{ndf} \approx 1$).
- Observables with largest A_C departure ($\delta p_T, p_n$) show inflated χ^2 consistently across all models, confirming a regularisation smearing artefact.
- Fake-data truth closure is unbiased ($\chi^2/\mathrm{ndf} \ll 1$) throughout.

Backup: generator predictions

- **Inclusive**: Per-observable $d\sigma/dx$ from GENIE (gst), GiBUU (`FinalEvents`), NEUT (neutvect), and NuWro (event output), flux-averaged to $10^{-38}\,\mathrm{cm}^2/\mathrm{Ar}$.
- **Proton-tagged W/TKI**: Reprocessed with identical proton tag and TKI reconstruction (NuWro and NEUT evaluated inside SL7 Apptainer container).
- **Rebinning**: FTE rebinned to analysis binning for direct overlay.